

Algebra 1
Unit 2
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

1. Pam was simplifying the expression and made an error. Her steps are shown. Find and fix the mistake.

$$\left(\frac{4x^3y^3 \cdot 2xy}{3xy^2}\right)^2$$

Step 1: $\left(\frac{8x^4y^4}{3xy^2}\right)^2$

Step 2: $\frac{64x^6y^6}{9x^2y^4}$

Step 3: $\frac{64x^4y^2}{9}$

Step 1: $\left(\frac{8x^4y^4}{3xy^2}\right)^2$

Step 2: $\frac{64x^8y^8}{9x^2y^4}$

Step 3: $\frac{64x^6y^4}{9}$

2. Write a note to Pam explaining her error.

When using the Power of a Power Exponent Law, multiply the exponent to all the powers inside the parentheses in both the numerator and denominator.

For questions 3 through 5, find an equivalent expression using the laws of exponents.

3. $\frac{14x^6y^6}{12x^{10}y^3}$

$$\frac{7y^3}{6x^4}$$

4. $\frac{4x^6y^6}{6x^2} \cdot \frac{3x}{4y}$

$$\frac{12x^7y^6}{24x^2y} = \frac{x^5y^5}{2}$$

5. $\left(\frac{4x^2y^3}{5x^0y^5}\right)^2$

$$\frac{16x^4y^6}{25y^{10}} = \frac{16x^4}{25y^4}$$

6. Prove $2^{2x+4} = 16 \cdot 2^{2x}$ by using the laws of exponents.

$$2^{2x} \cdot 2^4 = 2^{2x} \cdot 16 = 16 \cdot 2^{2x}$$

7. Prove $4^{x-2} = \frac{4^x}{16}$ by using the laws of exponents.

$$4^x \cdot 4^{-2} = \frac{4^x}{4^2} = \frac{4^x}{16}$$

8. Write an equivalent expression for $(-3.1)^{4x} \cdot (-3.1)^{2x+7}$

$$(-3.1)^{6x+7}$$

9. Use the law of exponents to write an equivalent expression for 5^{7-3x} .

$$\frac{78125}{5^{3x}}$$

10. Use the law of exponents to write an equivalent expression for $(2.8)^{6x+8}$.

$$3778.019983 \cdot (2.8)^{6x}$$